

1. Given the busy life of people nowadays, exercise is an essential activity for maintaining and improving health. Training at a gymnastic center does not consist of simply doing random exercises, but is planned based on a clear training strategy and effective training methods.

A gymnastic center introduced indexes to measure the progress of the training, but the data thus far have not been utilized effectively. Therefore, the gymnastic center decides to develop a gymnastic index management system to provide better services for its customers. Figure 1 shows an E-R diagram of the gymnastic index management system (incomplete), and Table 1 shows the description of each entity in Figure 1.

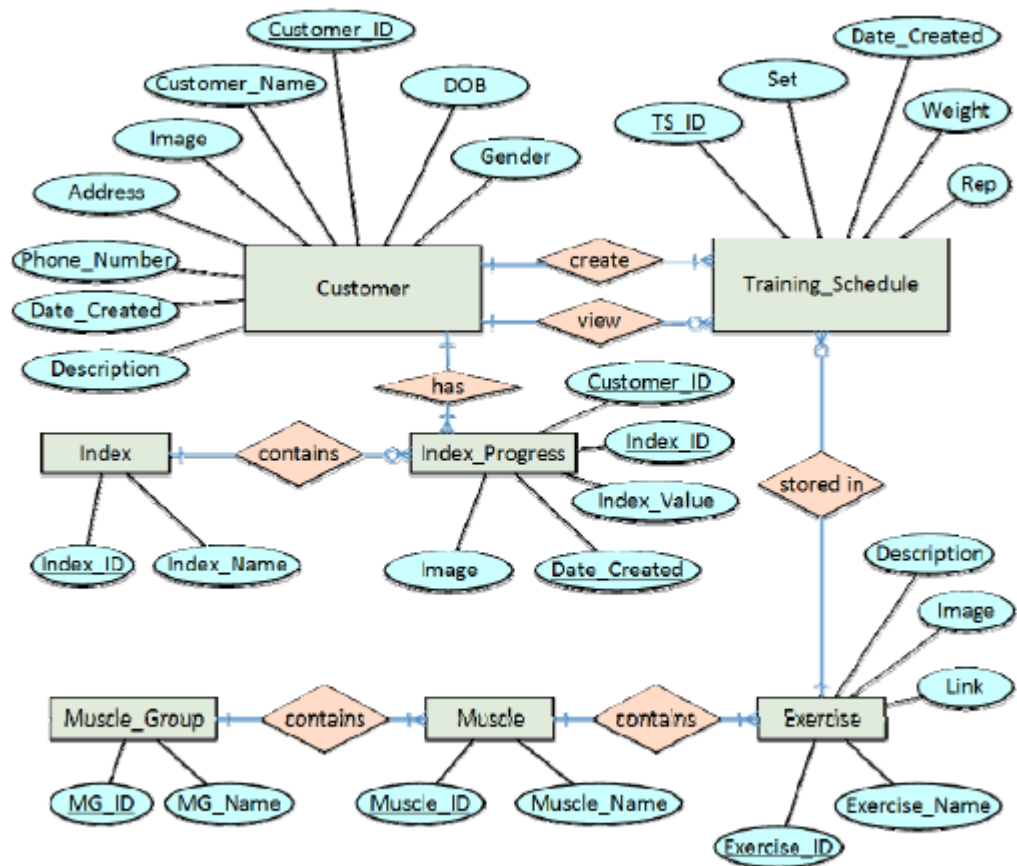


Figure 1. E-R diagram of the gymnastic index management system (incomplete)

Table 1. Description of each entity in Figure 1

Entity name	Description
Customer	Information about each registered customer.
Index	Name of each index.
Index_Progress	Index value and image of a customer at a difference time. A customer can monitor the changes in his/her index values and images.
Muscle_Group	Major muscle group in the human body. "Muscle Group" is abbreviated to "MG".
Muscle	Name of each muscle that belongs to the major muscle group.
Exercise	Information about each exercise that is associated with the muscle.
Training_Schedule	Training schedule of exercises for a customer. "Training Schedule" is abbreviated to "TS".

Subquestion 1

From the answer groups below, select the correct answer to be inserted in each blank in the following description.

Figure 1 is incomplete because it has four missing attributes which are primary keys.

In order to complete the E-R diagram, Figure 1 should be revised as follows:

- (1) Add the attribute to the entity Exercise.
- (2) Add the attribute to the entity Muscle.
- (3) Add the attributes Customer_ID and Exercise_ID to the entity .

Answer group for A and B

- | | |
|----------------|----------------|
| a) Customer_ID | b) Exercise_ID |
| c) Index_ID | d) MG_ID |
| e) Muscle_ID | f) TS_ID |

Answer group for C

- | | |
|-----------------|----------------------|
| a) Index | b) Index_Progress |
| c) Muscle_Group | d) Training_Schedule |

Subquestion 2

From the answer groups below, select the correct answer to be inserted in each blank in Figure 2.

Figure 2 shows an SQL statement that lists all the fitness index progress of a specific customer during the specified period from the beginning date to the end date.

Here, :FromDate and :ToDate are host variables that contain the beginning date and the end date, respectively, and :CustID is a host variable that contains the ID of the specific customer.

```
SELECT Customer.Customer_ID, Customer_Name, Index_Name, Index_Value
FROM   Customer, Index, Index_Progress
WHERE  Customer.Customer_ID = :CustID
      AND Customer.Customer_ID = Index_Progress.Customer_ID
      

```

Figure 2. SQL statement

Figure 3 shows an example of the output list. The list is arranged in ascending order of the date created.

Customer ID	Customer Name	Index Name	Index Value
C00001	Peter	Pectorals	7
C00001	Peter	Deltoids	5
C00001	Peter	Obliques	12
C00001	Peter	Deltoids	8

Figure 3. Example of the output list

Answer group for D

- a) AND Customer.Date_Created = Index_Progress.Date_Created
AND Index_Progress.Date_Created BETWEEN :FromDate AND :ToDate
- b) AND Customer.Date_Created = Index_Progress.Date_Created
AND Index_Progress.Date_Created IN (:FromDate, :ToDate)
- c) AND Index.Index_ID = Index_Progress.Index_ID
AND Index_Progress.Date_Created BETWEEN :FromDate AND :ToDate
- d) AND Index.Index_ID = Index_Progress.Index_ID
AND Index_Progress.Date_Created IN (:FromDate, :ToDate)

Answer group for E

- a) GROUP BY Index_Progress.Date_Created
- b) GROUP BY Index_Progress.Index_ID, Index_Progress.Date_Created
- c) ORDER BY Index_Progress.Date_Created
- d) ORDER BY Index_Progress.Index_ID, Index_Progress.Date_Created DESC